DIFFERENTIALS OF THE UNIVERSAL FIRST ORDER THICKENING OF AN ALGEBRAIC SPACE MORPHISM RESTRICT TO THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3336-3347 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3349-3363, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally unramified morphisms of algebraic spaces, universal first order thickenings, differentials, and etale localisation compatibility; the reduction is retained.

Source reference. Original lemma lemma-functoriality-differentials: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1062–1083. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-localize-differentials: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–996. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-universal-thickening-localize: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3309–3325. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-differentials-universally-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3363–3363. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $h : Z \to X$ be a formally unramified morphism of algebraic spaces over B. Let $Z \subset Z'$ be the universal first order thickening of Z over X with structure morphism $h' : Z' \to X$. The canonical map*

$$dh' : (h')^*\Omega_{X/B} \to \Omega_{Z'/B}$$

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

induces an isomorphism $h^*\Omega_{X/B} \to \Omega_{Z'/B} \otimes \mathcal{O}_Z$.

Proof. The map $c_{h'}$ is the map defined in Lemma lemma-functoriality-differentials. If $i : Z \to Z'$ is the given closed immersion, then $i^*c_{h'}$ is a map $h^*\Omega_{X/S} \to \Omega_{Z'/S} \otimes \mathcal{O}_Z$. Checking that it is an isomorphism reduces to the case of schemes by étale localization, see Lemma lemma-universal-thickening-localize and Lemma lemma-localize-differentials. In this case the result is More on Morphisms, Lemma more-morphisms-lemma-differentials-universally-unramified. □